

1. **How would you use shallow and deep tube wells for the water supply in the terai? Under what conditions each is used and why?**

Answer:

Shallow and deep tube wells are the important sources of water supply in Terai region due to abundant availability of groundwater resources. The selection of source depends upon the depth of aquifer, water demand, quality of groundwater and hydrogeological conditions. Studies from 2022–2025 reported a significant decline in groundwater levels in many Terai districts, with some locations reporting up to about 90% of shallow tube wells going dry seasonally. Degradation of Chure–Bhabar recharge areas (sand and gravel mining, deforestation, wetland loss) has sharply reduced natural recharge. It is found that about 79% of water sources in the Chure are drying, which directly threatens Terai aquifers that depend on this recharge zone. Also, as cited by DWSSM, over 50% of tested tube wells in some Terai belts had arsenic concentrations close to or above safe limits, posing serious public-health risks.

**Shallow tube wells:**

They are type of open well that rest on the top water bearing strata and draw their supplies from the surroundings.

**Conditions:**

- i. Used where the water table is shallow, generally within the depth of 15-50 m.
- ii. Used in the areas with the good quality of groundwater.
- iii. Used in small villages, individual households and small irrigation schemes.

**Advantages of shallow tube wells:**

- i. Shallow tube well has low construction and maintenance cost.
- ii. It is easy and quick to install.
- iii. Water from the shallow tube well can be pumped with small pumps or hand pumps.
- iv. Suitable for small water demand.

**Limitations:**

- i. Water level in shallow tube well fluctuates seasonally.
- ii. There is higher risk of contamination from surface pollutants.
- iii. Shallow tube well may dry up during prolonged dry period.
- iv. Groundwater extraction exceeding rainfall recharge.

**Deep tube wells:**

They are the type of open well that rests on impervious strata and draws their supplies from the previous formation lying below the impervious strata through the borehole constructed in impervious strata.

**Conditions:**

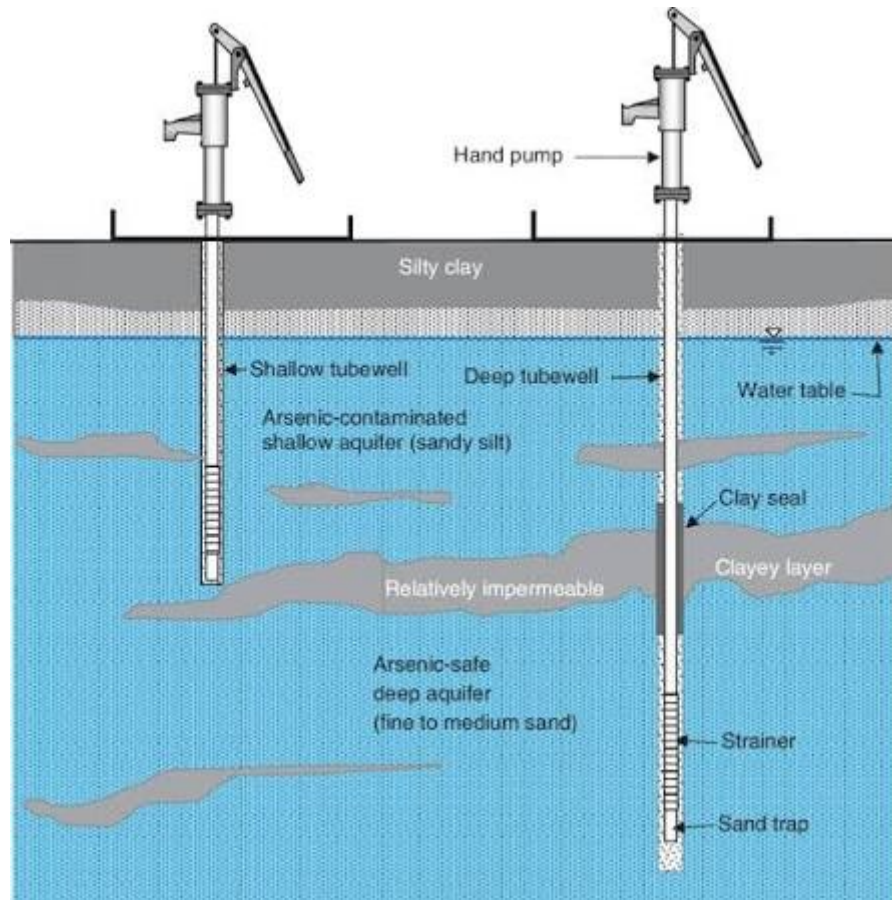
- i. Used where the shallow aquifers are contaminated or inadequate.
- ii. Used where large quantities of water are required.
- iii. Used in urban areas, towns and large irrigation project.
- iv. Used where groundwater occurs in confined aquifers present at larger depths.

**Advantages:**

- i. Deep tube wells provide a reliable year-round water supply.
- ii. It provides better protection from surface contamination.
- iii. Deep tube well is capable of supplying required number of large discharges.
- iv. It is suitable for municipal and large-scale irrigation systems.

**Limitations:**

- i. Deep tube well requires high drilling and installation cost.
- ii. It requires skilled operation and maintenance.



2. A village in mild western development region of Nepal has a design year population of 500 per capita demand recommended for that particular village is 65 liter per day. The demand is to be met by a continuous system of supply from a spring source with the safe yield of 0.5 lps. The consumption pattern is:

| Time (hr) | Consumption (%) |
|-----------|-----------------|
| 5-7       | 25              |
| 7-12      | 35              |
| 12-17     | 20              |
| 17-19     | 20              |
| 19-5      | 0               |

**Is balancing storage tank storage necessary? Calculate its capacity if necessary.**

Answer:

Given:

Design population,  $P=500$

Per capita demand = 65 L/person/day

Safe yield of spring = 0.5 L/s

Supply system = Continuous (24 hours)

Step 1: Determine total daily water demand

$$Q_d = P \times q = 500 \times 65 = 32500 \text{ L / day}$$

$$Q_d = 32.5 \text{ m}^3 / \text{day}$$

Step 2: Check if the source is adequate.

$$\text{Safe yield of spring} = 0.5 \text{ L/s}$$

$$\text{Daily yield} = 0.5 \times 86400 = 43200 \text{ L/day} = 43.2 \text{ m}^3/\text{day}$$

Since

$$43.2 > 32.5, \text{ the source is adequate.}$$

Step 3: Continuous supply rate

$$= \frac{32500}{24} = 1354.17 \text{ L/hr}$$

Step 4: Hourly demand

i. 5-7 hr (2hr)

$$\text{Demand} = 25\% \times 32500 = 8125 \text{ L}$$

$$\text{Hourly demand} = \frac{8125}{2} = 4062.5 \text{ L/hr}$$

$$\text{Net withdrawal \& storage} = 4062.5 - 1354.17 = 2708.33 \text{ L/hr}$$

$$\text{For 2 hr} = 2708.33 \times 2 = 5416.66 \text{ L}$$

$$\text{Storage required} = 5416.66 \text{ L}$$

ii. 7-12 hr (5 hrs)

$$\text{Demand} = 35\% \times 32500 = 11375 \text{ L}$$

$$\text{Hourly demand} = \frac{11375}{5} = 2275 \text{ L/hr}$$

$$\text{Deficit} = 2275 - 1354.17 = 920.83 \text{ L/hr}$$

$$\text{For 5 hr} = 920.83 \times 5 = 4604.17 \text{ L}$$

$$\text{Maximum deficit} = 5416.66 + 4604.17 = 10020.83 \text{ L}$$

iii. 12-17 hr (5 hr)

$$\text{Demand} = 20\% \times 32500 = 6500 \text{ L}$$

$$\text{Hourly demand} = 1300 \text{ L/hr}$$

$$\text{Supply} = 1354.17 \text{ L/hr}$$

$$\text{Surplus} = 54.17 \text{ L/hr}$$

$$\text{For 5 hr} = 54.17 \times 5 = 270.85 \text{ L}$$

$$\text{Storage remaining} = 10020.83 - 270.85 = 9749.98 \text{ L}$$

iv. 17-19 hr (2 hr)

$$\text{Demand} = 6500 \text{ L}$$

$$\text{Hourly demand} = 3250 \text{ L/hr}$$

$$\text{Deficit} = 3250 - 1354.17 = 1895.83 \text{ L/hr}$$

$$\text{For 2 hr} = 1895.83 \times 2 = 3791.66 \text{ L}$$

$$\text{Maximum storage deficit} = 9749.98 + 3791.66 = 13541.64 \text{ L}$$

v. 19-5 hr (10 hr)

$$\text{Demand} = 0$$

Supply continues

$$\text{Storage filled} = 1354.17 \times 10 = 13541.7 \text{ L}$$

Hence storage returns  $\hat{=}$  full.

Step 5: Capacity of Balancing Storage Tank

$$\text{Maximum storage required} = 13542 \text{ L} \approx 13.5 \text{ m}^3$$

Hence, balancing storage is necessary and its capacity is  $13.5 \text{ m}^3$

### 3. What do you mean by treatment of water and why is it necessary? What do you mean by flocculation?

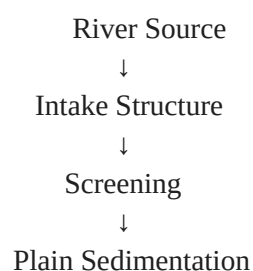
Answer:

**Water treatment** is the process of removing physical, chemical, and biological impurities from raw water to make it safe and suitable for drinking and other domestic uses. Sundarjal water treatment plant in Kathmandu valley is one of the major water treatment plants in Nepal. Almost all major towns with piped supply (e.g., Dhulikhel, Pokhara, Bharatpur, Butwal, Nepalgunj) have smaller conventional WTPs, serving surface-water schemes.

**Necessity of water treatment:**

- i. **Remove Suspended and Floating Matter**
  - Water treatment helps to eliminate sand, silt, clay, algae, and other suspended impurities.
- ii. **Remove Disease-Causing Microorganisms**
  - It also helps in destroying bacteria, viruses, protozoa, and other pathogens to prevent water-borne diseases.
- iii. **Improve Taste, Odor, and Color**
  - It also removes objectionable color, unpleasant taste, and foul odor to make water aesthetically acceptable.
- iv. **Reduce Turbidity**
  - It produces clear and transparent water by removing fine suspended particles.
- v. **Provide Safe and Potable Water**
  - It ensures the treated water meets national and international drinking water quality standards (e.g., WHO guidelines).

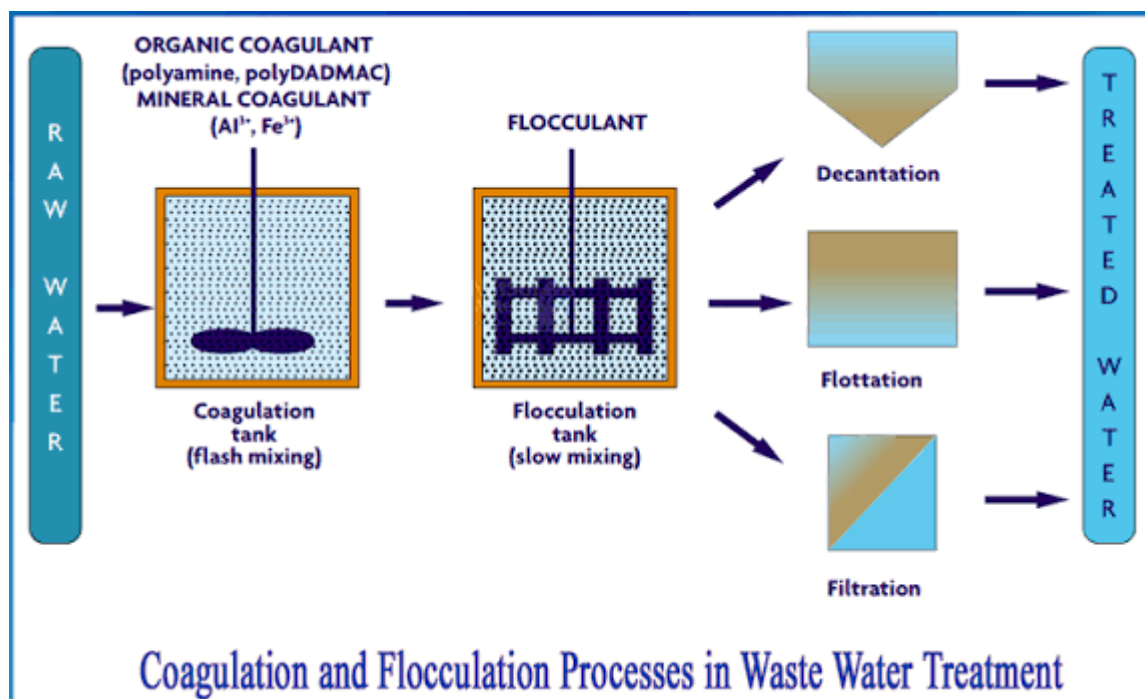
**Schematic flow diagram illustrating the possible treatment units involved in the water treatment process:**



↓  
Sedimentation with Coagulation  
↓  
Filtration  
↓  
Disinfection (Chlorination)  
↓  
Aeration  
↓  
Softening  
↓  
Miscellaneous

### Flocculation:

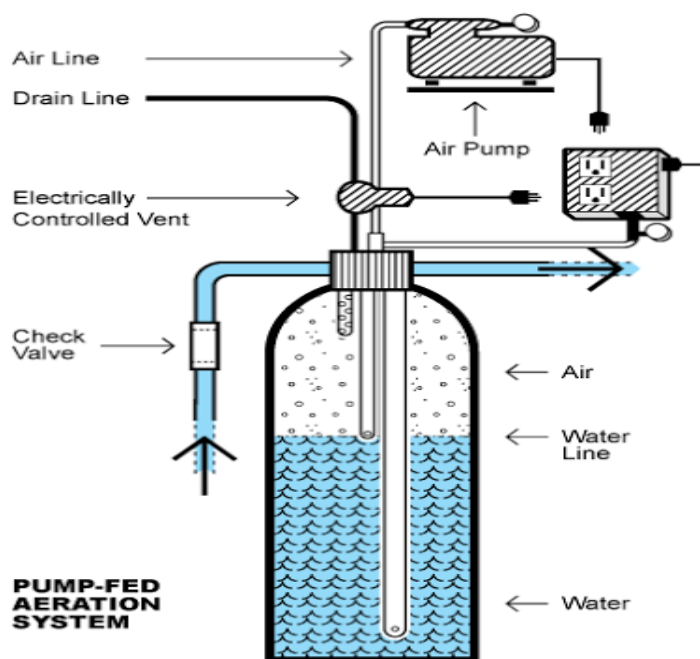
- Flocculation is the process of formation of floc. The zones of floc are positively charged and they react with negatively charged particles to form heavier floc, which settles down under gravitational force.
- The main purpose of coagulation-flocculation process is to remove very fine suspended and colloidal particles and reduce load in subsequent treatment plants.
- After thorough mixing of coagulants in the water in the next operation is flocculation.
- Flocculators are slow stirring mechanism which forms flocs. It mainly consists of paddles which are revolving at very slow speed about 2-3 rpm.
- The paddles may revolve on a vertical or horizontal shaft.
- The flocculators provide number of gentle contacts between the flocculating particles which are necessary for the successful formation of floc.
- The detention time for the best results is 30-60 minutes.
- The floc which has been formed above is allowed to settle and is separated from the water is done by keeping the water in sedimentation tanks known as coagulation basins.



4. **Why aeration is used in water treatment plants? Is it more commonly used with ground water or surface water and why? Give reasons**

Answer:

- The process of bringing water in contact with air to remove dissolved gasses present in water is known as aeration.
- Aeration removes  $\text{CO}_2$  gasses up to 70% due to which corrosion corrosiveness of water is reduced.
- Aeration is used in water treatment plants because it:
  - i. **Removes Dissolved Gases**
    - a. Removes carbon dioxide ( $\text{CO}_2$ ), hydrogen sulphide ( $\text{H}_2\text{S}$ ), methane ( $\text{CH}_4$ ), and other objectionable gases.
    - b. Eliminates bad taste and foul odor.
  - ii. **Oxidizes Iron and Manganese**
    - a. Converts dissolved ferrous iron ( $\text{Fe}^{2+}$ ) and manganese ( $\text{Mn}^{2+}$ ) into insoluble forms.
    - b. These precipitates can then be removed by sedimentation and filtration.
  - iii. **Increases Dissolved Oxygen (DO)**
    - a. Adds oxygen to water, improving its freshness and making it suitable for aquatic life and further treatment.
  - iv. **Removes Taste and Odor**
    - a. Eliminates unpleasant taste and smell caused by dissolved gases and organic compounds.
  - v. **Reduces Corrosiveness**
    - a. Removes excess carbon dioxide, thereby reducing the corrosive nature of water and protecting pipelines and equipment.
  - vi. **Removes Volatile Organic Compounds**
    - a. Helps strip some volatile chemicals and gases from water.
  - vii. **Improves Efficiency of Treatment**
    - a. Enhances the effectiveness of subsequent treatment processes such as coagulation, filtration, and disinfection.



Aeration is more commonly used for groundwater than for surface water. The reasons are listed below:

**i. Removal of dissolved gases:**

- Groundwater often contains dissolved gases such as:
  - Carbon dioxide (CO<sub>2</sub>)
  - Hydrogen sulphide (H<sub>2</sub>S)
  - Methane (CH<sub>4</sub>)
- Aeration removes these gases, improving the taste and odor of water.

**ii. Oxidation of Iron and Manganese:**

- Groundwater frequently contains dissolved iron (Fe<sup>2+</sup>) and manganese (Mn<sup>2+</sup>).
- Aeration oxidizes them into insoluble forms, which can then be removed by sedimentation and filtration.

**iii. Increase in dissolved oxygen:**

- Groundwater generally has low dissolved oxygen (DO) because it has little contact with air.
- Aeration increases the DO content, improving water quality.

**iv. Reduction of corrosiveness:**

- Groundwater often contains excess carbon dioxide, making it corrosive.
- Aeration removes CO<sub>2</sub> and reduces corrosion of pipes and treatment equipment.

**v. Better taste and odor:**

- Groundwater may have unpleasant taste and odor due to dissolved gases.
- Aeration improves its aesthetic quality.

**5. Write short notes on:**

**a. Impounding reservoirs**

**b. Break point chlorination:**

**c. Artesian well**

Answer:

**a. Impounding reservoirs:**

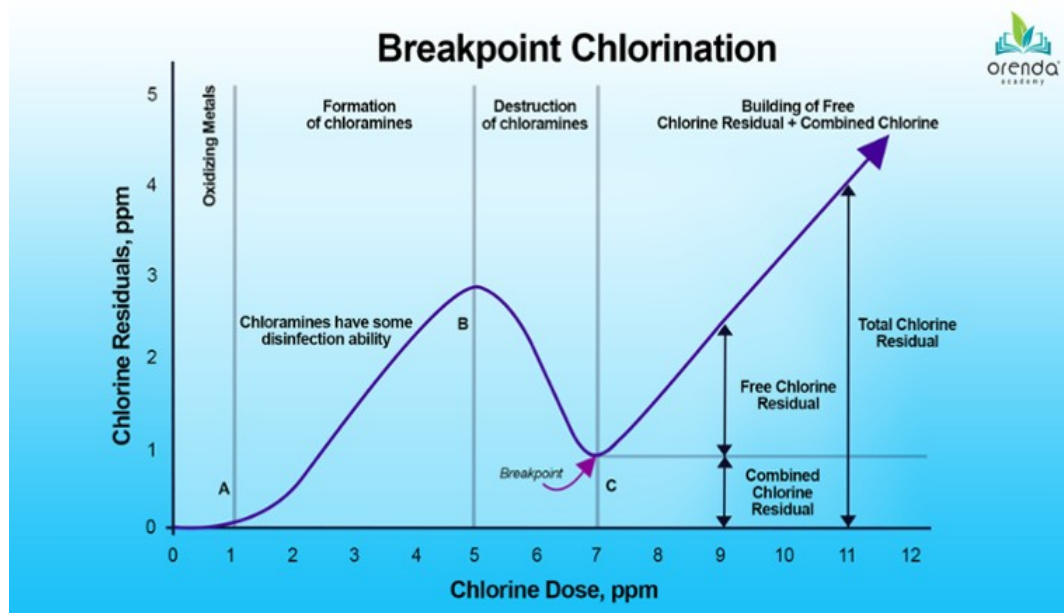
- Impounded reservoirs are the reservoirs formed by constructing dams, weir or bund across the valleys cut by streams.
- The quality of water in a storage reservoir mainly depends on the quality of river water and needs to be properly analyzed and treated before applying to the consumers.
- Impounded reservoirs are generally constructed when the river has not sufficient discharge to meet the water demand in summer season.
- Indra sarowar, Dhap dam, upper seti reservoir are some of the pioneer impounding reservoirs constructed in Nepal.
- During the site selection of impounded reservoir, following points should be noted.
  - i. The site should have narrow river channel.
  - ii. The site should be located at higher elevation than that of treatment plant and distribution area.
  - iii. The site should have availability of sufficient quantity of water.
  - iv. Important infrastructures should not get submerged during construction reservoir.
  - v. watershed should be free from swampy area and bed soil should not contain impurities.
  - vi. topography should be favorable for site and should have sufficient construction materials.





### b. Break point chlorination:

- When the chlorine is added to the water, it reacts with organic and inorganic matter to form common compounds. Some portion of chlorine remains as residual chlorine.
- Now, if the chlorine dose is increased, the combined available residual chlorine also increases. When the chlorine dose is so increased, the compounds get oxidized and the substances which are newly formed do not react with ortho-chloride to show any residual.
- If the addition of chlorine is continued and a graph is plotted between chlorine dose and residual chlorine, a curve is obtained as shown.



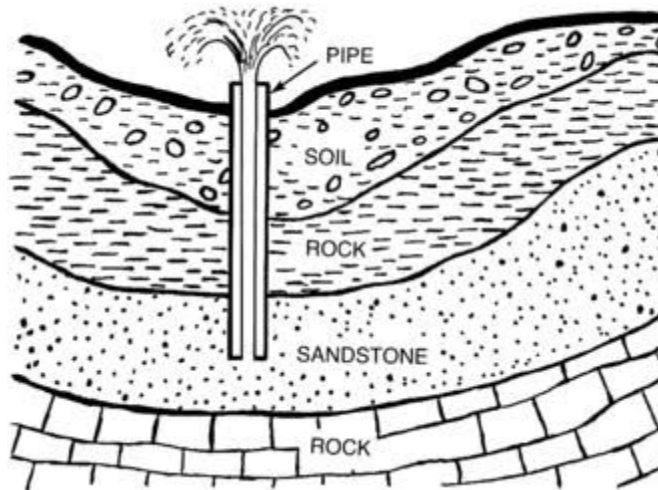
- On studying the curve, we can notice that the residual chlorine in the beginning increases with the applied chlorine dose, but after point B, it suddenly drops up to point C and then increases. Portion OB shows the formation of chloramines and portion BC shows their oxidation.
- Point C at which residual chlorine again starts increasing is known as break-point chlorination. After reaching this point, if chlorine is further added, it remains as free residual chlorine and curve becomes straight line.
- Advantages of break-point chlorination:
  - It completely oxidizes ammonia and other impurities of water.
  - The color of water due to organic matters is also reduced.
  - It completely destroys all the disease bacteria.



It prevents growth of weeds in the water.

**c. Artesian well:**

- Those wells in which water rises due to its pressure above the level at which it is encountered in the aquifer is known as artesian wells.
- These are caused when previous stratum is enclosed between the two impervious strata with the outcrop so high above the ground that the hydraulic line is above the ground level.
- This is usually formed in the valley.
- Flowing artesian wells have been documented on the eastern bank of the Koshi River in the eastern Terai, where confined aquifers receive recharge from upland areas and the river system.
- Artesian wells in Nepal are found almost exclusively in the Terai (Inner and Outer Terai) sedimentary basins in the districts such as Jhapa, Morang, Sunsari, Saptari, Siraha, Dhanusha, Mahottari, Sarlahi, Rautahat, Bara, and Parsa, where geological conditions favor confined aquifers.
- When the boring is done, water comes out without requiring to be pumped when the hydraulic gradient line lies above the top level of well.
- When it goes below the top level of the well, pumping is necessary.
- The quality of water from artesian well is the same as that of deep well water.



**6. What are the relative advantage and disadvantage of separate and combined sewer system? What are the design considerations for the two types of sewer systems?**

Answer:

**Separate system:**

Separate system consists of two different sewers, one for convergence of sewage and another for rainwater.

It requires large areas since cleaning is not possible in small sewers.

If the rainfall is heavy and for short period of time and if sanitary sewage is required to be pumped, then separate system is provided.

**Advantages:**

- i. Load on treatment plant is reduced because waste type is taken separately.
- ii. Uniform process for purification.
- iii. Reduces the possibility of pollution due to overflow during excess rainfall.
- iv. Rainwater can be directly discharged into river.

**Disadvantages:**

- i. Requires two set of sewers, hence costly.
- ii. Occupies large area.
- iii. Choking problem can be seen in separate system.
- iv. There is a possibility of wrong connection due to confusion.

**Combined system:**

Combined system consists of only one sewer to carry both sanitary sewage and storm water.

Sewage must be treated before discharged into the streams.

It is suitable if street washings are as impure as sewage and must be treated before discharging.

**Advantages:**

- i. Combined system reduces the cost as it requires only one sewer.
- ii. It also reduces maintenance cost.
- iii. Strength of sewage is reduced by dilution.
- iv. House plumbing is easy and economic.
- v. Self-cleansing velocity of sewer can be easily achieved in combined sewerage system.

**Disadvantages:**

- i. Large dimensions of sewer are required in combined system.
- ii. Ventilation problem may occur.
- iii. Load on treatment plant increases in combined system.
- iv. Overflow of sewage during rainy season can occur in combined system.
- v. Handling and transportation of large size of sewer is difficult.

Design considerations for two types of sewer systems:

**i. Separate system:**

- Sanitary sewage is designed based on population, per capita water demand, infiltration, and peak factor. Storm sewer is designed separately using rainfall-runoff analysis.
- Sewer sizes for separate system is smaller; storm sewers sized independently.
- Maintain self-cleansing velocity (about **0.6–0.75 m/s**) to prevent silting;
- Limit maximum velocity (about **3 m/s**) to avoid scouring.
- Treatment plant designed only for sanitary sewage.
- Provide adequate slope for gravity flow and easy maintenance.
- Manholes are installed at changes in direction, gradient, junctions, and at regular intervals.
- Separate networks require higher initial cost but allow easier future expansion and better treatment.

**ii. Combined system:**

- Combined system is designed for the combined flow of sanitary sewage and stormwater during the design storm.
- Sewer size is large sewers required to accommodate both sewage and storm runoff.
- Combined system must safely carry peak combined flow while maintaining self-cleansing velocity during dry weather.
- Treatment plant or overflow structures must account for variable combined flows.
- Manholes are designed to withstand higher peak flows.
- Combined Sewer Overflow structures may be necessary to prevent flooding during heavy rainfall.
- Lower initial cost but higher operation and maintenance costs due to larger flow variations.

**7. Explain aerobic and anaerobic decomposition of sewage.**

Answer:

**Aerobic decomposition:**

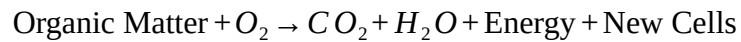
Aerobic decomposition is the process in which aerobic bacteria decompose organic matter in the presence of dissolved oxygen (DO).

Aerobic bacteria use oxygen to oxidize organic matter.

Organic matter is converted into stable end products such as carbon dioxide (CO<sub>2</sub>), water (H<sub>2</sub>O), nitrates (NO<sub>3</sub><sup>-</sup>), sulphates (SO<sub>4</sub><sup>2-</sup>) etc.

### Characteristics:

- Requires dissolved oxygen.
- Produces less offensive odor.
- Decomposition is rapid.
- Produces stable and harmless end products.
- Common in activated sludge process, trickling filters, and oxidation ponds.



### Anerobic decomposition:

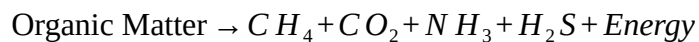
Anaerobic decomposition is the breakdown of organic matter by anaerobic bacteria in the absence of oxygen.

Anaerobic bacteria digest organic matter without oxygen.

The process occurs in enclosed tanks or septic environments and forms the stable products such as methane ( $\text{CH}_4$ ), Carbon dioxide ( $\text{CO}_2$ ), Hydrogen sulphide ( $\text{H}_2\text{S}$ ), Ammonia ( $\text{NH}_3$ ) and Stable sludge.

### Characteristics:

- No dissolved oxygen is required.
- Produces foul odor due to  $\text{H}_2\text{S}$  and other gases.
- Decomposition is slower than aerobic decomposition.
- Generates biogas (methane), which can be used as fuel.
- Common in septic tanks, Imhoff tanks, and anaerobic digesters.



## 8. What is oxidation pond? How does it function? What data do you require for its design?

Answer:

Oxidation pond is an open pond, flowing through earthen basins, designed and constructed for the treatment of sewage and industrial waters.

Detention period in oxidation pond is longer ranging from few to several days.

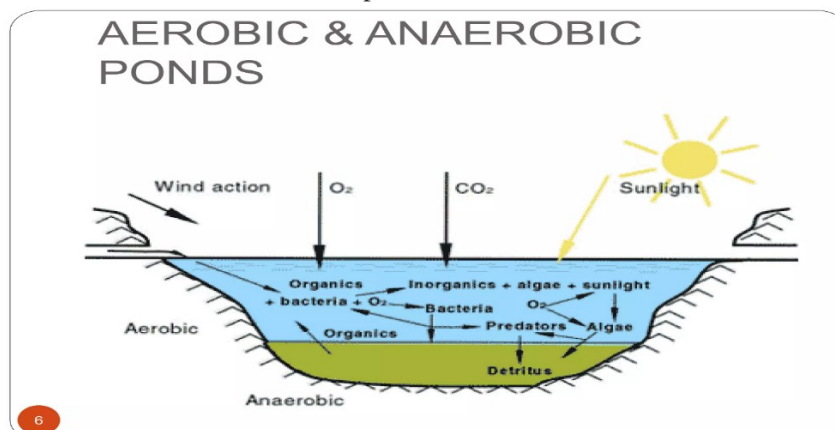
Oxidation pond is used to treat municipal sewage as well as biodegradable industrial sewage.

Stabilization of sewage is done by aerobic, anaerobic as well as facultative bacteria in oxidation pond. There are three zones created in oxidation pond. They are:

- Aerobic zone
- Anaerobic zone
- facultative zone

### Functions of oxidation pond:

- To settle the organic matter at the bottom of the pond.
- To reduce coliform present in the sewage.
- To reduce the load on the treatment plant.



For the design of oxidation pond, the data we require are listed below:

- i. Pond shape:  
There should not be any corners or extended portion.  
Shape may be round, square or any regular shape with the length not exceeding 3 times the width.
- ii. Embankment:  
Top width=1.5-3 m'  
Freeboard= 0.5 m (minimum)  
Inner slope= 1:1 to 1.5:1  
Outer slope= 2:1 to 2.5:1
- iii. Detention period= 10-45 days
- iv. Depth of oxidation pond= 1-1.5 m
- v. Surface area < 0.5 hectares. If > 0.5 hectares, multiple ponds can be constructed.
- vi. Loading criteria:  
For cold region= 55 kg/ha/ day  
For tropic region= 300 kg/ha/day

9. **What do you know about the physical, chemical and biological characteristics of sewage?**

Answer:

The properties of sewage that determines the quality and helps in design of sewerage systems and treatment plants are known as characteristics of sewage. There are three characteristics of sewage:

i. **Physical characteristics:**

The physical characteristics of sewage are:

a. **Color:**

- Fresh domestic sewage has light brown in color.
- Due to aerobic decomposition, the color changes into light to medium grey.
- After time passes due to anaerobic decomposition, color change into dark grey or black.

b. **Odor:**

- Fresh domestic sewage is odorless.
- Due to anaerobic decomposition,  $H_2S$  gas is formed which emits the smell of rotten egg.

c. **Temperature:**

- Temperature increases the biological activity of bacteria up to 60 degrees and then it decreases.
- When the temperature of sewage is more, the content of dissolved oxygen and other gases becomes less.

d. **Turbidity:**

- Turbidity in sewage is due to total solid contents.
  - a. colloidal solids:
  - b. Suspended solids
  - c. Dissolved solids:

ii. **Chemical characteristics:**

The chemical characteristics of sewage are:

a. **pH:**

- Fresh domestic sewage is alkaline in nature. i.e., 7.3 to 7.5.
- As time passes, pH falls due to product of acids by bacterial action and sewage tends to be acidic.

b. **Organic and inorganic solids:**

- Organic solids compose of carbon, hydrogen and oxygen, produced from plant and animal.
- Organic solids are completely volatilized from water at higher temperature of 550°C for 20 minutes.
- Inorganic solids composed of nitrogen, chlorine, ammonia in large proportions.

**c. Nitrogen content:**

- Nitrogen and phosphorous are essential elements required for growth of micro-organisms, plants and animals.
- Removal or reduction of nitrogen in waste water helps to reduce algae growth.

**iii. Biological characteristics:**

Different micro-organisms such as sewage bacteria, protozoa, fungi, algae etc. contribute for the decay of organic matter present in the sewage.

**a. Bacteria:**

There are three types of bacteria according to the oxygen requirement:

Aerobic bacteria: requires free oxygen for their survival.

Anaerobic bacteria: survive and flourish in absence of free oxygen.

Facultative bacteria: can survive and flourish with or without free oxygen

10. **a. What do you understand by greenhouse effect? What are the causes? How can it be mitigated?**  
**b. List in details the physical, biological and socio-economic baseline information that have to be collected during EIA study of a water supply project.**

Answer:

Greenhouse effect is the condition resulted when the certain gases in the earth's atmosphere trap outgoing infrared radiation which keeps the earth warm enough to support life. However, an increase in greenhouse gases due to human activities enhances this effect, leading to global challenges like global warming and climate change. The major greenhouse gases are: Carbon dioxide (CO<sub>2</sub>), Methane (CH<sub>4</sub>), Nitrous oxide (N<sub>2</sub>O), Water vapor (H<sub>2</sub>O), Ozone (O<sub>3</sub>), Chlorofluorocarbons (CFCs) and other fluorinated gases.

**Causes of greenhouse effect:**

The greenhouse effect is mainly caused by an increase in greenhouse gases (GHGs) in the atmosphere.

1. **Burning of Fossil Fuels** such as coal, petroleum, and natural gas used in industries, vehicles, and power plants release carbon dioxide (CO<sub>2</sub>).
2. **Deforestation:** Cutting down forests reduces the absorption of CO<sub>2</sub>, increasing its concentration in the atmosphere.
3. **Industrial Activities:** Factories emit CO<sub>2</sub>, methane (CH<sub>4</sub>), nitrous oxide (N<sub>2</sub>O), and fluorinated gases.
4. **Agriculture:** Rice cultivation and livestock produce methane (CH<sub>4</sub>). Also, fertilizers release nitrous oxide (N<sub>2</sub>O).
5. **Waste Disposal:** Decomposition of organic waste in landfills releases methane.
6. **Use of Refrigerants:** CFCs, HCFCs, and HFCs used in refrigeration and air conditioning act as potent greenhouse gases.

**Effects:**

- Increase in global average temperature (global warming)
- Melting of glaciers and polar ice caps
- Rise in sea level
- Changes in rainfall patterns
- More frequent floods, droughts, heat waves, and storms
- Loss of biodiversity and ecosystem imbalance
- Reduced agricultural productivity in many regions

Mitigation measures:

- a. **Reduce Fossil Fuel Use:** Promoting renewable energy such as solar, wind, and hydropower.
- b. **Afforestation and Reforestation:** Plant more trees and protect existing forests to absorb CO<sub>2</sub>.
- c. **Improve Energy Efficiency:** Use energy-efficient appliances, buildings, and industrial processes.
- d. **Sustainable Transportation:** Encourage public transport, cycling, walking, and electric vehicles.
- e. **Control Industrial Emissions:** Adopt cleaner technologies and emission-control systems.
- f. **Proper Waste Management:** Reduce, reuse, recycle, compost organic waste, and capture landfill methane.
- g. **Climate-Friendly Agriculture:** Improving fertilizer management and livestock practices to reduce methane and nitrous oxide emissions.
- h. **International Cooperation:** Implementation of global climate agreements and policies to reduce greenhouse gas emissions.

Based on the **Environment Protection Act, 2076**, the collection of detailed baseline information is required across physical, biological, and socio-economic systems to assess the existing environmental status.

The following detailed baseline information must be collected during an EIA study for a water supply project:

**i. Physical and chemical baseline information:**

- a. **Geography and Topography:**
  - Detailed mapping of the project's geographical location, surface conditions, land use, and land capacity.
- b. **Climate and Meteorology:**
  - Historical data on rainfall, temperature, humidity, and wind direction for the project area.
- c. **Hydrology and Watershed:**
  - Identification of rivers, lakes, and other water sources; assessment of watershed status, flood history, and risks such as Glacial Lake Outburst Floods (GLOF).
- d. **Geology and Soil:**
  - Study of underground rock structures, soil types, and geological formations; assessment of risks related to landslides, soil erosion, and seismic (earthquake) risks.
- e. **Quality Indicators:**
  - **Air Quality:** Measurement of PM<sub>10</sub>, PM<sub>2.5</sub>, Carbon Monoxide, and Total Suspended Particles (TSP).
  - **Water Quality:** Analysis of Total Suspended Solids (TSS), Dissolved Oxygen (DO), Conductivity, Total Nitrate, pH levels, E. coli, and Arsenic levels.
  - **Noise Levels:** Monitoring of equivalent noise levels during both day and night.

**ii. Biological baseline information:**

- a. **Forest Resources:**
  - List of forest types (based on climate and vegetation) and their distance from the project site, including forest management status.
- b. **Flora (Vegetation):**
  - Identification of plant species by local and scientific names; classification of their conservation status according to the IUCN Red List, CITES, and the Government of Nepal.
- c. **Fauna (Wildlife):**
  - Detailed listing of mammals, birds, reptiles, and fish in the area, including their scientific names, families, and conservation/protection status.



d. **Habitats:**

- Description of natural habitats and the communities of species residing within them.

e. **Non-Timber Forest Products (NTFPs):**

- Identification and usage patterns of local non-timber resources.

**iii. Socio-economic and cultural baseline information:**

a. **Demographics:**

- Identification of affected settlements, household numbers, average household size, male-to-female ratios, and the ethnic or religious composition of the community.

b. **Economic Status:**

- Information on primary income sources, including agriculture (crop types, production levels, soil quality, and chemical fertilizer use), animal husbandry, employment levels, and migration patterns.

c. **Water Usage:**

- Current status of upstream and downstream water use, specifically for agricultural, domestic, or industrial purposes.

d. **Culture and Heritage:**

- Mapping of historical and cultural heritage sites, religious monuments, open spaces of cultural significance, and traditional beliefs or customs.

e. **Language and Traditions:**

- Documentation of local mother tongues, major festivals, ceremonies, and traditional practices, including cremation sites and rituals.

f. **Infrastructure and Services:**

- Existing status of educational institutions (schools), healthcare facilities (hospitals/health posts), communication services (telephone, internet, TV), and transportation networks (roads/access paths).

g. **Project-Specific Details:**

- For water supply, information on existing water sources, waste management systems, and any sensitive landmarks near the proposed project site.